Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/18/2023
PI Name:	Nemec-Bakk, Ashley Ph.D.		
Project Title:	The Use of Two New Ground-Based Models of Deep Space Travel to Study the Role of Mitochondria and Oxidative Stress in Cardiovascular Effects		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	72205-7199	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH-RFA-2201-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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COI Name (Institution):	Boerma, Marjan Ph.D. (MENTOR: University of Arkansas, Little Rock)		
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Task Description:	POSTDOCTORAL FELLOWSHIP On future missions into deep space, astronauts will be subject to chronic low doses of ionizing radiation. Since only a small number of astronauts have traveled outside of low-Earth orbit, it is difficult to estimate the physiological risk of long distance space travel on individual organ systems. Therefore, studies determining the effects of space radiation rely on animal models to estimate the risks of degenerative health effects and to understand biological mechanisms involved. This project addresses three knowledge gaps in this area of research: 1) There is a lack of studies that investigate the combination of galactic cosmic radiation (GCR) and microgravity on the cardiovascular system and risks for degenerative diseases; 2) Biological and pathological pathways of degenerative cardiovascular diseases due to space radiation and microgravity are not yet fully understood; and 3) Although there has been an increase in woman astronauts since 1950, most ground-based studies have been performed on males. To address these knowledge gaps, this project will make use of two ground-based models in which male and female C57BL/6J mice will be subjected to the modeled deep space environment with two methods that are being developed in our laboratory. The first model makes use of the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory to create a mixed ion field simulating GCR. The second model makes use of a chronic gamma ray facility at our home institution to expose mice to chronic low-dose rate ionizing radiation for one month. In both models, we will combine radiation exposure with simultaneous hind limb unloading (HLU) to model fluid shifts in microgravity, thereby simulating the environment in deep space. The project tests the hypothesis that exposure to simulated GCR or chronic low-dose rate exposures to ionizing radiation will result in mitochondrial damage and increased oxidative stress, which will have an adverse impact on the function of the cardiovascular system. This hypothesis will be tested in the following aims: Aim 1: Determine the effects of simulated GCR and HLU on mitochondria, oxidative stress, and function of the heart, major arteries, and microvasculature in a mouse model. The influence of sex on cardiovascular outcomes of simulated GCR and HLU will be assessed. Aim 2: Test the effectiveness of secoisolariciresinol diglucoside, a natural antioxidant, as a potential countermeasure against cardiovascular effects of chronic low-dose gamma ray exposure and HLU. Cardiac function will be measured with high-resolution echocardiography, and the eye structure and retinal vasculature will be imaged with optical coherence tomography. Arterial function will be measured with ex vivo perfused aorta obtained from the mice at the end of the studies. Lastly, tissue samples will be collected to measure mitochondrial function and antioxidant enzyme activities. Taken together, these hypothesis-driven aims will fill crucial knowledge gaps that are relevant to NASA’s future long duration space flights beyond low-Earth orbit. This study will elucidate if and how GCR and microgravity can initiate cardiovascular degenerative diseases and whether there is a different risk between males and females.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Research activities are progressing through project year 1. Near-term research impacts are the development of primary cell types from the natural hibernator model, the golden mantled ground squirrel. We have developed protocols and cells from brain, heart, skeletal muscle, liver, and kidney over the past year. These tools will be available for completion of the project's aims, but also provide research tools for future investigations. We have also obtained long-range genomic sequencing data from this species, which will become publicly available once the genome assembly is complete. The original project objectives are to use a parallel approach to investigate control of transcription and translation over a range of “hibernation” temperatures in a natural hibernator in vivo, alongside human cells and tissues in vitro. First, to determine whether homeostatic control is maintained in a natural model in cool, intermediate, and hot temperatures, we hibernated golden mantled ground squirrels at 4, 12, 25, and 30°C to elicit different body temperatures. In addition to data on oxygen consumption (respirometry) and torpor patterns (body temperature telemetry) to determine efficacy of hibernation, we have submitted samples for RNAseq to measure differential gene expression (from liver, heart, kidney, and brain) during torpor. These data will be analyzed in the upcoming project year. Second, to identify the optimum temperature for human cells and organoids in vitro that best maintains homeostatic control, we proposed to use experimental cooling to induce metabolic depression. We will also quantify oxygen consumption reduction and mitochondrial function as indices of metabolic depression at 4, 12, 25, and 30°C and in parallel, evaluate transcription and translation rates by tracking incorporation of radioactive labels as well as sequencing. If human cells cannot effectively maintain homeostasis at low temperatures, then either intervention is required, or high temperature hibernation is necessary. We have acquired human cells and piloted these experiments over the past project year, and will complete activities in the coming year. In our third aim, we will trial an intervention that promotes Internal Ribosome Entry Sites/IRES-mediated protein synthesis and track indicators of protein synthesis (at ribosomes). In the coming year, we will evaluate transcripts poised for synthesis in cultured cells from ground squirrels exposed to experimental temperatures, and compare metrics of cell health and survival in human cells with and without rapamycin intervention. NASA seeks innovation to reversibly manipulate metabolic and homeostatic processes on demand as a key spaceflight countermeasure, maximizing crew resilience and health, thus reducing mission risk. Temperature is a likely contributor to any strategy that will ultimately reduce metabolic processes. This project will examine the limits of temperature regulation to elicit metabolic depression in humans and natural hibernators and determine the tradeoffs between energy savings and regulation of homeostasis across varied temperatures, focusing on control of protein synthesis. This approach will take critical steps forward in developing temperature regulation (hypothermia) as well as stimulation of IRES-mediated protein synthesis as a countermeasure.
Bibliography Type:	Description: (Last Updated: 11/06/2025)

Articles in Other Journals or Periodicals	Nemec-Bakk A, Boerma M. "How does space exploration affect the heart and blood vessels?" Front Young Minds. 2023 Dec 7;11:1232651. https://doi.org/10.3389/frym.2023.1232651 , Dec-2023
Articles in Peer-reviewed Journals	Nemec-Bakk AS, Sridharan V, Desai P, Landes RD, Hart B, Allen AR, Boerma M. "Effects of simulated 5-ion galactic cosmic radiation on function and structure of the mouse heart." Life. 2023 Mar 15;13(3):795. https://doi.org/10.3390/life13030795 ; PMID: 36983950; PMCID: PMC10057791 , Mar-2023
Awards	Nemec-Bakk A. "International Congress for Radiation Research (ICRR) Travel Award, May 2023." May-2023